Wage-Price Spirals: The Contract Accelerator

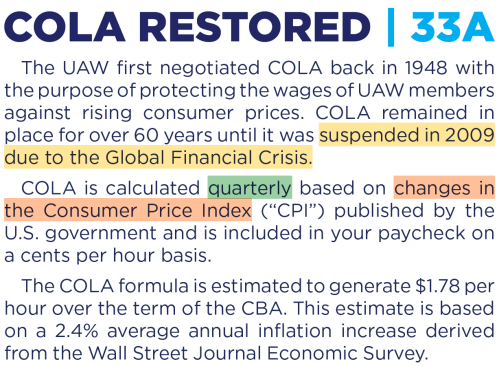
Xilin Yuan

Yale University

October 8, 2026

Wage Setting Around the Recent Inflation Surge

(a) Cost-of-living adjustment (COLA) clauses allow inflation-contingent wages



COLA RESTORED | 33A

The UAW first negotiated COLA back in 1948 with the purpose of protecting the wages of UAW members against rising consumer prices. COLA remained in place for over 60 years until it was suspended in 2009 due to the Global Financial Crisis.

COLA is calculated quarterly based on changes in the Consumer Price Index ("CPI") published by the U.S. government and is included in your paycheck on a cents per hour basis.

The COLA formula is estimated to generate $1.78 per hour over the term of the CBA. This estimate is based on a 2.4% average annual inflation increase derived from the Wall Street Journal Economic Survey.

Notes: Panel (a) is taken from a summary report of the UAW–Ford collective bargaining agreement signed in 2022.

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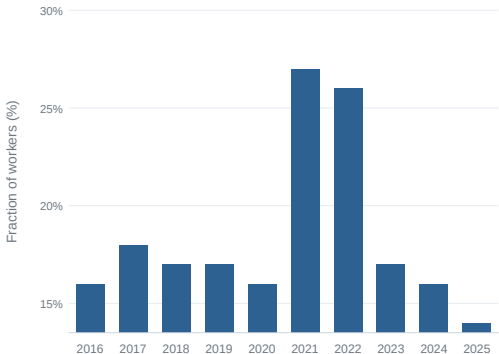
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(b) Share of U.S. job stayers receiving more than one base-wage change in a year



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Research Questions

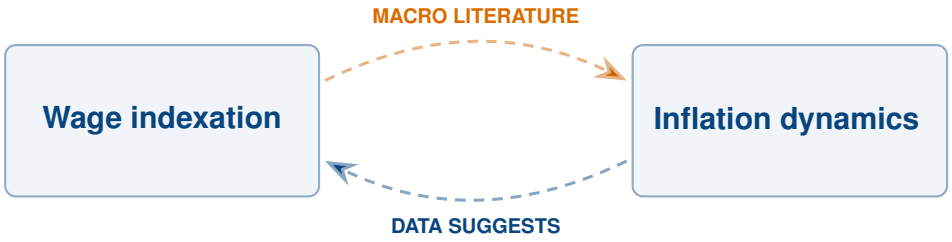
Wage indexation

Inflation dynamics

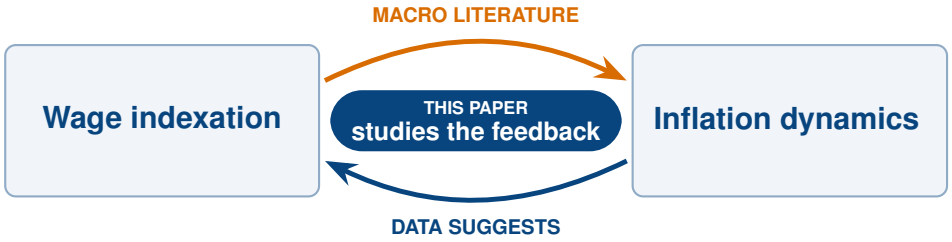
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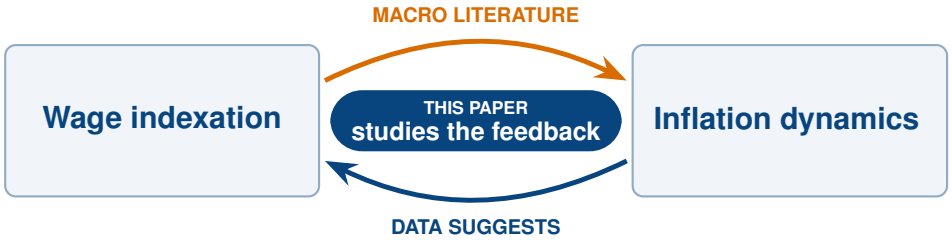
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Research Questions



Research Questions



- Can the feedback mechanism generate wage-price spirals—and in what sense?
- Can monetary policy stabilize inflation by reshaping the equilibrium contract choice?

Results I: Endogenous Contracts and Two Layers of Spirals

- Motivating facts from a new dataset at the labor-contract level:
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2. The contract accelerator: contract choices are **strategic complements**
   - higher inflation volatility→more COLA adoption→even higher inflation volatility

Results II: Policy, Model Validation, and Quantification

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- Quantification: the contract accelerator in the Great Inflation and the Great Moderation
  - amplifies major inflation spikes (1973-75, 79-81) by **14%**, compared to holding COLA fixed
  - **doubles** the contribution of Volcker-type monetary policy to inflation stabilization

Related Literature

Wage indexation as insurance against inflation

Azariadis, 1978; Blanchard, 1979; Azariadis and Cooper, 1985; Ehrenberg et al., 1983; Hendricks and Kahn, 1985; Card, 1983, 1986; Hurst et al., 2026; Guerreiro et al. 2026; Afrouzi et al., 2024; Gödl et al. 2026; Moscarini and Postel-Vinay, 2023; Pilossoph and Ryngaert, 2024.

- This project: provides measures of indexation in a broad panel

Wage indexation and inflation propagation

Barro 1976; Gray, 1976; Fischer, 1977, 1983; Erceg et al., 2000; Christiano et al., 2005; Smets and Wouters, 2007; Blanchard, 1986; Lorenzoni and Werning, 2023; Blanchard, 1979; Carrillo et al., 2022; Alexanian, 2025.

- This project: endogenizes both the prevalence and strength of indexation

Uncertainty and coordination

Bloom 2009; Bloom et al., 2018; Jurado et al., 2015; Cooper and John, 1988; Benhabib and Farmer, 1994; Morris and Shin, 2003; Angeletos and La'O, 2010, 2013.

- This project: connects these two through a concrete contract choice margin

Plan

1. Data and Motivating Evidence
2. Model Setup
3. Analyzing the Model
4. Model Validation
5. Calibration and Quantitative Exercise

Data and Motivating Evidence

Data and Background Information

Major Collective Bargaining Settlements (1968–1995)

- Systematic measurement
- FOIA release; now public
- 21,543 contracts, 3,918 firm–union pairs

▸ Contract example ▸ Summary statistics ▸ Scheduled example ▸ COLA example ▸ COLA by sector ▸ Substitution

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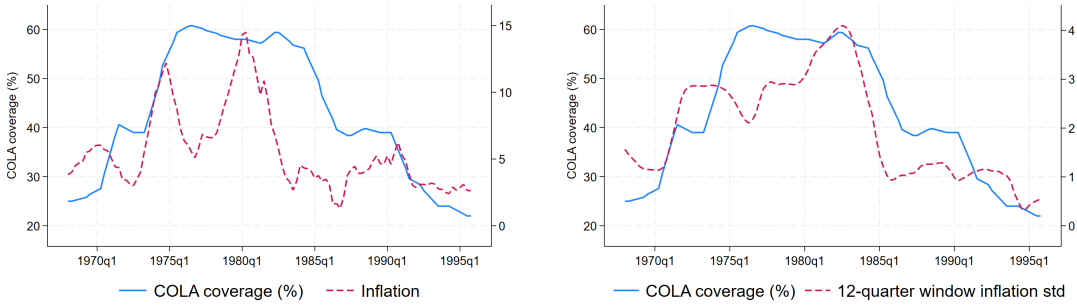
react to inflation surprises

- Revenue-contingent contracts: Rare and limited to a few giant auto firms

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COLA Coverage Comoves with Inflation and Level Volatility

Figure: COLA coverage vs. inflation (left) and inflation volatility (right)

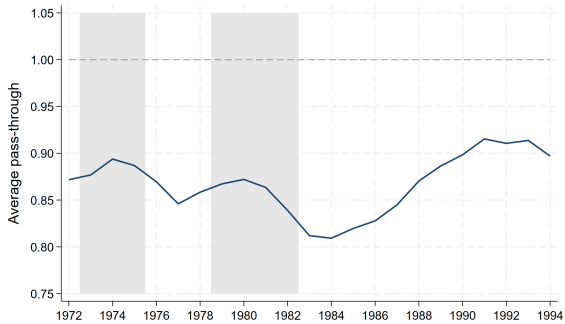


Notes: COLA coverage is the fraction of workers with COLA provision among all workers covered by major collective bargaining agreements in the US. Inflation on the left panel is quarterly inflation, and inflation volatility on the right panel is the standard deviation of inflation from the 12-quarter rolling window.

▸ post 2000 ▸ other countries ▸ aggregate pass-through

COLA Pass-through is High but Still Incomplete

Figure: Average price pass-through to nominal wages in *new* COLA contracts each year



Notes: COLA pass-through is defined as the percentage change in nominal hourly wage in response to a one-percentage increase in the price level implied by the COLA formula. The average is taken over all newly signed contracts with a COLA clause each year. The time series is smoothed using the moving average method on a three-year rolling window. The shaded area indicates the two inflation spikes.

▸ active COLA ▸ cross-section

Model Setup

Setup: Upstream Service Firms

- Differentiated services enter a CES service bundle

$$X_t = \left(\int_0^1 X_{it}^{\frac{\varepsilon_x - 1}{\varepsilon_x}} di \right)^{\frac{\varepsilon_x}{\varepsilon_x - 1}} \quad \Longrightarrow \quad P_{it}^X = \left(\frac{X_{it}}{X_t} \right)^{-1/\varepsilon_x} P_t^X$$

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- CES demand and technology determine operating profit

$$\begin{aligned} q_{it} &= \frac{P_{it}^X}{P_t} X_{it} - \frac{W_{it}}{P_t} L_{it} \\ &= \zeta_t \times (\eta_{it} L_{it})^{1 - 1/\varepsilon_x} - w_{it} L_{it} \end{aligned}$$

where $\zeta_t := \frac{P_t^X}{P_t} X_t^{1/\varepsilon_x}$ is the real revenue factor which the firm takes as given

Firm Risk and External Financing Costs

$$q_{it} = \zeta_t \times (\eta_{it} L_{it})^{1-1/\varepsilon_x} - w_{it} L_{it}$$

- Firm chooses labor L_{it} under uncertainty about idiosyncratic TFP shock η_{it}
  - low η_{it} can imply negative profit
  - more likely if w_{it} high or ζ_t low
- Negative profit requires equity financing with real cost $\Phi(q_{it})$ Gilchrist et al., 2017

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- Firm value and labor demand

$$Q(\zeta_t, w_{it}) := \max_{L_{it}} \mathbb{E}^{\eta} \left[q_{it} - \Phi(q_{it}) \mathbb{I}(q_{it} < 0) \right] \implies L^*(\zeta_t, w_{it})$$

$\implies$ Firms dislike paying high wages especially in low-revenue states

Environment: Firm–Union Wage Contracting

Firm-specific Union

fixed worker membership
risk-averse, hand-to-mouth

Taylor Contract

fixed N-period duration
renewal cohorts staggered

Objective Function

maximize joint value
two-sided commitment

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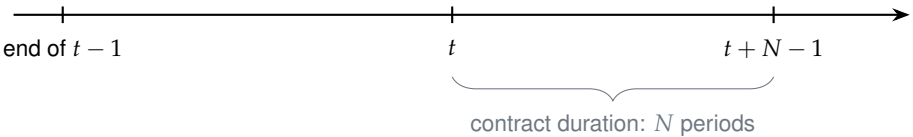
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Renewal

choose contract

$\{W_{t,t-1}(\cdot), ..., W_{t+N-1,t-1}(\cdot)\}$

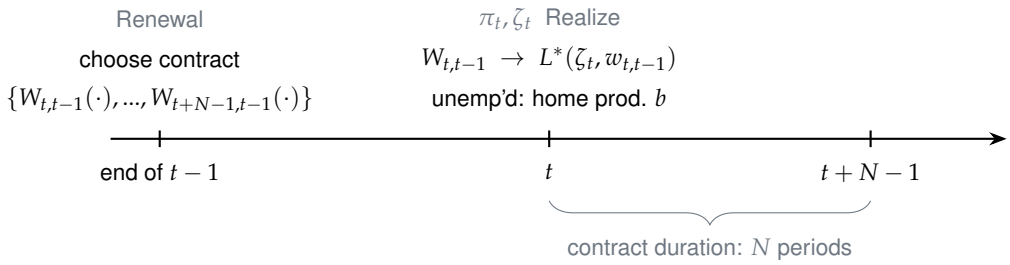


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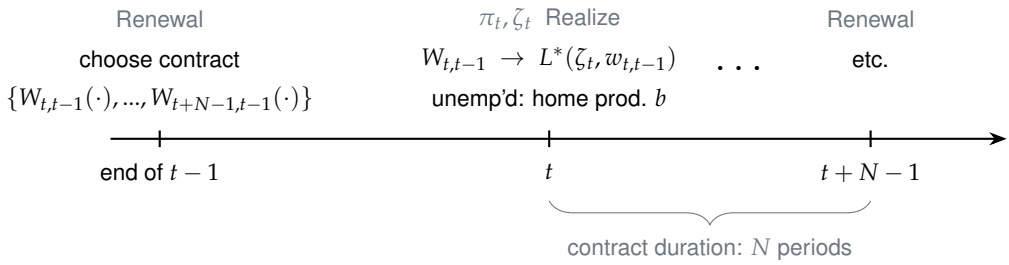


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Feasible Contracts Differ Only in State Contingency

Non-contingent class $\mathcal{N}$

$$\{W_{s,t-1}\}_{s=t}^{t+N-1}$$

N wage numbers chosen at $t-1$

based on expected inflation

Inflation-contingent class $\mathcal{C}$

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- Contingent contracts carry a cognitive calculation cost $\chi_i \sim F_\chi$ Martínez-Marquina et al. (2019)
- Two roles of inflation
  - affects workers' purchasing power
  - contains info. about the real revenue factor ζ_t

The Contracting Problem

- Optimal COLA: N nominal wage functions of concurrent inflation

$$V_{t-1}^{\mathcal{C}} = \max_{\{W_s(\pi^s)\}\in\mathcal{C}} \sum_{s=t}^{t+N-1} \beta^s \mathbb{E}_{t-1} \left[\underbrace{L^*\left(\zeta_s, \frac{W_s(\pi^s)}{P_s}\right) u\left(\frac{W_s(\pi^s)}{P_s}\right) + \left[\bar{L} - L^*\left(\zeta_s, \frac{W_s(\pi^s)}{P_s}\right)\right] u(b)}_{\text{union preference: utilitarian}} + \underbrace{\lambda Q\left(\zeta_s, \frac{W_s(\pi^s)}{P_s}\right)}_{\text{firm value}} \right]$$

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- Adoption margin: compare the net value from COLA and from non-contingent contracts

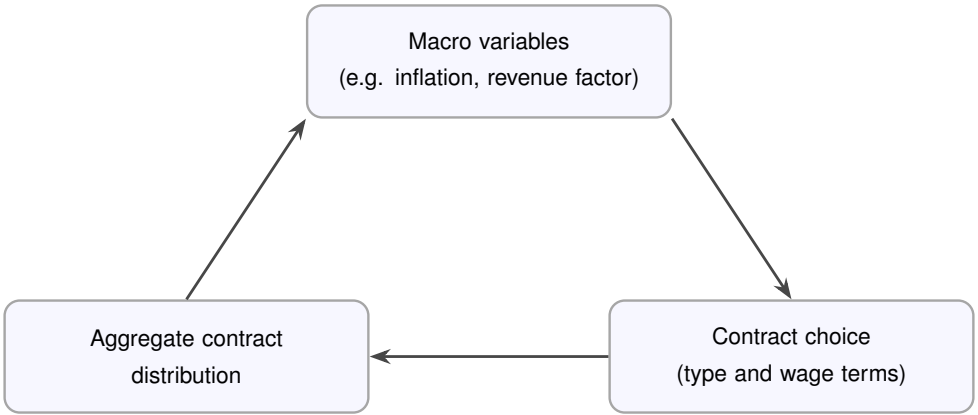
$$\max\left\{V_{t-1}^{\mathcal{C}} - \chi_i, V_{t-1}^{\mathcal{N}}\right\}$$

Environment: Other Standard NK Blocks

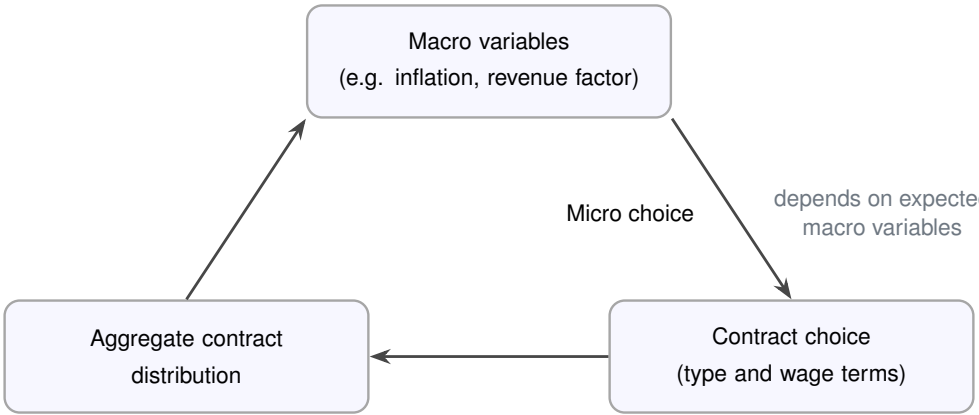
- A continuum of monopolistically competitive downstream firms $f \in [0,1]$
  - linear production using service bundle from upstream: $Y_{ft} = X_{ft}$
  - aggregate into final good $Y_t = \left(\int_0^1 Y_{ft}^{\frac{\varepsilon_{yt}-1}{\varepsilon_{yt}}} df \right)^{\frac{\varepsilon_{yt}}{\varepsilon_{yt}-1}} \implies$ markup shock: $u_t := -\frac{1}{\varepsilon_y - 1} \hat{\varepsilon}_{yt}$
  - Calvo friction: reset price with prob. $1-\theta$
- Representative entrepreneur with discount factor $\beta e^{\rho_t} \implies$ demand shock: ρ_t
- Central bank sets a Taylor-type policy rule

$$i_t = i_0 + \phi_\pi \pi_t + \phi_y \hat{y}_t$$

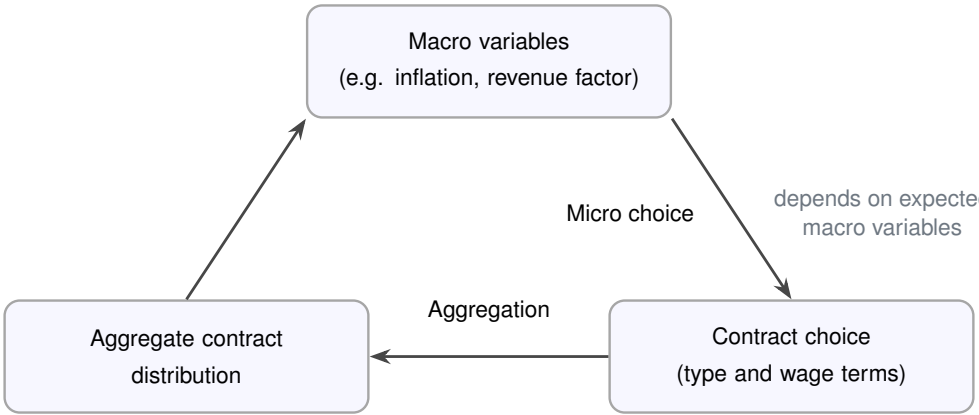
The Structure of the Model



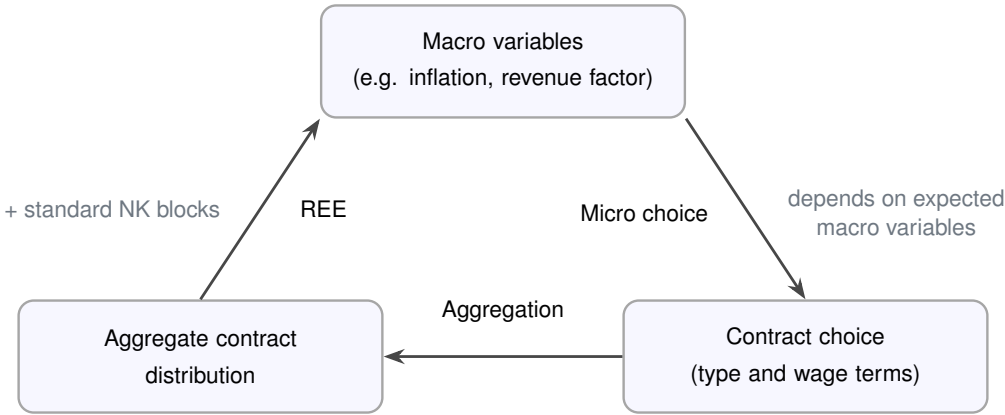
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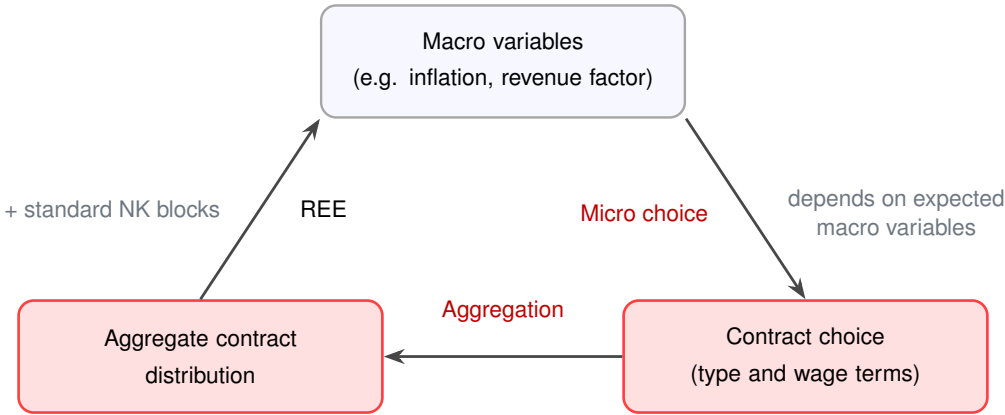
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The Structure of the Model



The Structure of the Model



Analyzing the Model

Four Hurdles: Two Standard, Two New

Standard Complications

1. Forward-looking NK block

agents forecast future contract adoption

2. Large state space

contract distribution and wage terms

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$\mathbb{E}_{t-1}[\zeta_t \mid \pi_t]$ needs eq'm joint d.g.p.

COLA wages feed back into that d.g.p.

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Removing Standard Hurdles to Isolate New Channels

Assumption 1

(1) Markup shock u_t and demand shock ρ_t are serially independent Gaussian with std σ_u and σ_ρ;
(2) one-period contracts: $N = 1$.

- Shocks i.i.d.: shuts down forward- and backward-looking complications
- One-period contracts: contract distribution represented by COLA share m_t
- Log-linearize the system *except* for m_t: aggregate state (u_t, ρ_t, m_t)

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- Conjecture the endogenous d.g.p. of $\hat{\zeta}_t$ and π_t for contract choice–how?

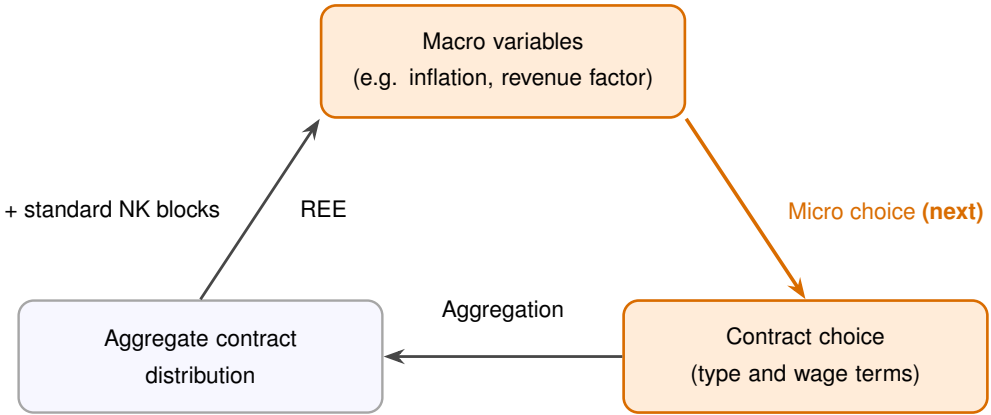
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- Conjecture the endogenous d.g.p. of $\hat{\zeta}_t$ and π_t for contract choice–how?

$$\hat{\zeta}_t = \zeta_\rho(m_t)\rho_t + \zeta_u(m_t)u_t$$
$$\pi_t = \pi_\rho(m_t)\rho_t + \pi_u(m_t)u_t$$

Micro Choice: Contract Adoption and Optimal Wage Terms



Risk Sharing Can Explain Partial COLA Pass-Through

Lemma 1

The realized real wages following the optimal wage terms are

$$\hat{w}_t^{\mathcal{C}} = \mathbb{E}_{t-1}[\hat{w}_t^* \mid \pi_t]$$

where $\hat{w}_t^* = \mu\hat{\zeta}_t$ would be the optimal real wage if $\hat{\zeta}_t$ were contractible, and $\mu > 0$ is a constant.

- Optimal wage term $\Longleftrightarrow$ linear projection of $\hat{w}_t^*$ on the available information

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$$\hat{w}_t^{\mathcal{C}} = \mathbb{E}_{t-1}\left[\hat{w}_t^*\right] + \frac{\mathrm{Cov}\left(\mu\hat{\zeta}_t, \pi_t\right)}{\mathrm{Var}\left(\pi_t\right)}\left(\pi_t - \mathbb{E}_{t-1}\left[\pi_t\right]\right) = \delta_t \pi_t$$

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- $\mathrm{Cov}\left(\hat{\zeta}_t, \pi_t\right) < 0 \implies$ optimal COLA pass-through is incomplete: $1 + \delta_t < 1$

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- $\mathrm{Cov}\left(\hat{\zeta}_t, \pi_t\right) < 0 \implies$ optimal COLA pass-through is incomplete: $1 + \delta_t < 1$
- Full pass-through is costly for the firm when inflation signals low revenues

Lemma 2

Given the conjecture $\pi_u, \zeta_u, \pi_\rho, \zeta_\rho$, the gain from COLA is given by

$$\mathcal{V}_{t-1} \propto [1 + \delta_t]^2 \operatorname{Var}_{t-1}(\pi_t)$$

where $\mathcal{V}_{t-1} := V^{\mathcal{C}}_{t-1} - V^{\mathcal{N}}_{t-1}$

- The gain captures the risk-sharing effect of COLA
- Individual cutoff: choose COLA if the gain $\mathcal{V}_{t-1}$ outweighs the adoption cost χ_i

Lemma 2

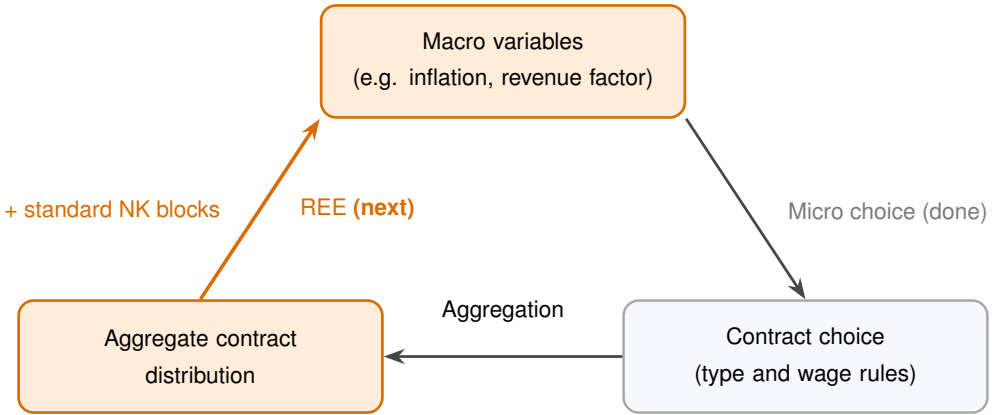
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- The gain captures the risk-sharing effect of COLA
- Individual cutoff: choose COLA if the gain $\mathcal{V}_{t-1}$ outweighs the adoption cost χ_i
- Expected COLA share m_t^e will summarize the GE feedback

Inner Fixed Pt: Conditional GE Given Contract Composition



One Statistic Carries Contracts into General Equilibrium

$$\widehat{\omega}_t = m_t \widehat{w}_t^{\mathcal{C}} + (1 - m_t) \widehat{w}_t^{\mathcal{N}}$$

$$\widehat{w}_t^{\mathcal{C}} = \delta_t \pi_t \qquad \widehat{w}_t^{\mathcal{N}} = -\pi_t$$

▸ K_t details ▸ NK details ▸ Regularity conditions ▸ REE Solutions

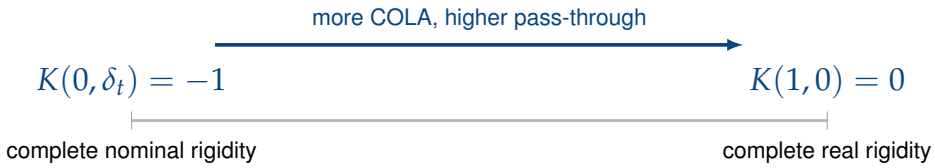
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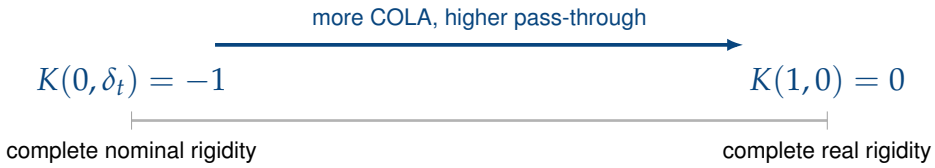
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One Statistic Carries Contracts into General Equilibrium

$$\widehat{\omega}_t = \underbrace{\left[m_t\delta_t - (1-m_t)\right]}_{K(m_t,\delta_t)}\pi_t$$



- Given m_t, equilibrium $\mathcal{K}(m_t) := K(m_t, \delta(m_t))$

▸ K_t details ▸ NK details ▸ Regularity conditions ▸ REE Solutions

COLA Affects the NKPC through the Conditional Wage-Price Spiral

Proposition 1

Given Assumption 1, for any m_t, the system admits a three-equation representation

$$NKPC: \pi_t = \frac{\kappa}{1 - \kappa \frac{\iota_2}{\iota_1} \mathcal{K}(m_t)} \left[\frac{\varepsilon_x - \iota_1}{\iota_1 \varepsilon_x} \hat{y}_t + u_t \right]$$

$$DIS: \hat{y}_t = \frac{\iota_1}{\iota_1 - \varphi_\zeta} \left[-\frac{\bar{C}^e}{\bar{Y}} \gamma (i_t - \rho_t) + \left(\varphi_\omega + \frac{\iota_2}{\iota_1} \varphi_\zeta \right) \mathcal{K}(m_t) \pi_t \right]$$

$$MP: i_t = \phi_\pi \pi_t + \phi_y \hat{y}_t$$

Under suitable conditions, $\mathcal{K}(m_t) < 0$ and increases with m_t.

▸ Conditions

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$$m_t \uparrow \implies \mathcal{K}(m_t) \uparrow$$

A higher COLA share steepens the NKPC

▸ Conditions

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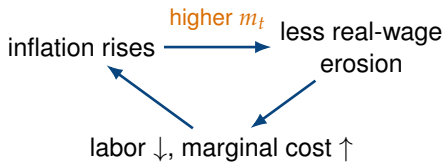
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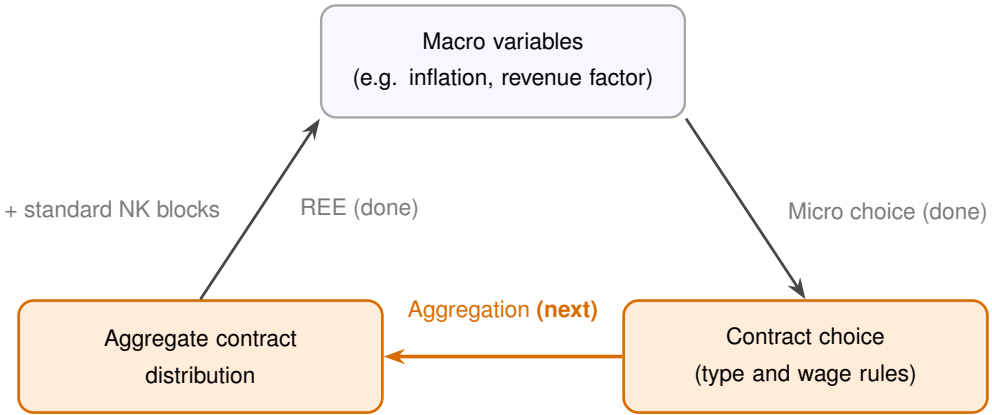
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▸ Conditions

Outer Fixed Point: General Contracting Equilibrium



Proposition 2

Under suitable conditions, the COLA gain increases with the expected COLA share:

$$\frac{\partial \mathcal{V}(m_t^e)}{\partial m_t^e} > 0$$

▸ Conditions

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- $m_t^e \uparrow \longrightarrow$ perceived inflation propagation $\uparrow$ $\quad \mathrm{Var}(\pi_t | m_t^e) \uparrow \quad \mathcal{V}(m_t^e) \uparrow$ own COLA adoption $\uparrow$

▸ Conditions

Strategic Complementarity through the Contract Accelerator

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- $m_t^e \uparrow$ → perceived inflation propagation ↑ → $\mathrm{Var}(\pi_t | m_t^e) \uparrow$ → $\mathcal{V}(m_t^e) \uparrow$ → own COLA adoption ↑
- Expecting more COLA makes COLA more attractive

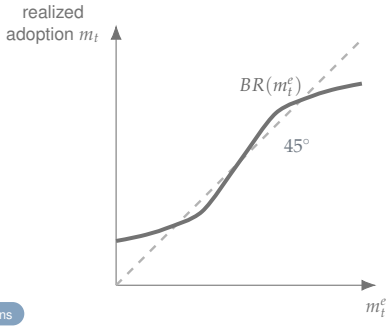
▸ Conditions

Strategic Complementarity Can Support Multiple Contract Regimes

Proposition 3

Under suitable conditions, there exists an equilibrium in which the COLA share m_t is deterministic and

$$m_t = F_\chi\left(\mathcal{V}(m_t^e)\right) := BR(m_t^e)$$
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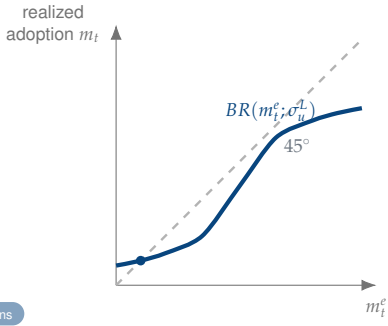
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Low uncertainty

unique low-COLA regime

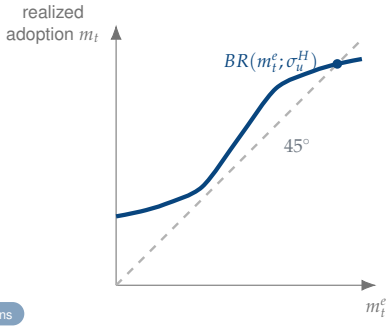
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High uncertainty

unique high-COLA regime

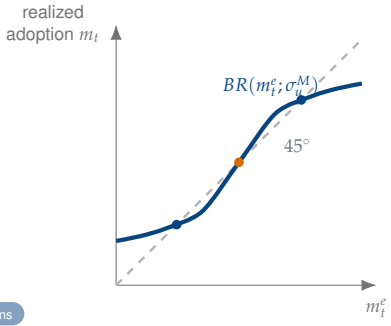
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Intermediate uncertainty

low, middle, and high regimes

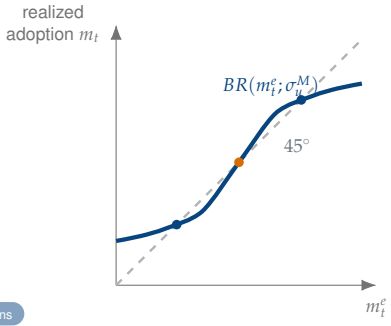
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▸ Conditions

Intermediate uncertainty

low, middle, and high regimes

Def: An equilibrium m_t^* is locally stable if

$$BR'(m_t^*) < 1$$

Role for monetary policy? (**next**)

Monetary Policy Stabilizes Inflation Twice

Proposition 4

Under suitable conditions, a stronger policy response to inflation ϕ_π lowers a locally stable m_t^*:

$$\frac{dm_t^*}{d\phi_\pi} < 0$$

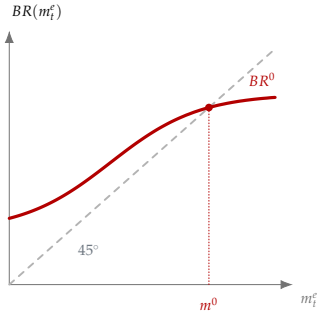
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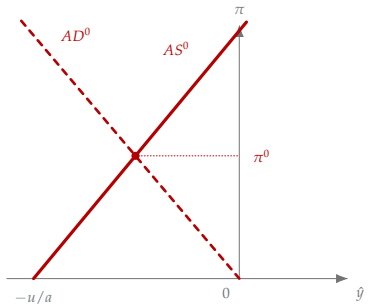
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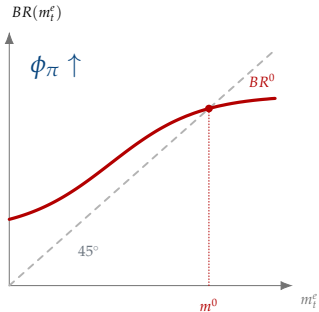
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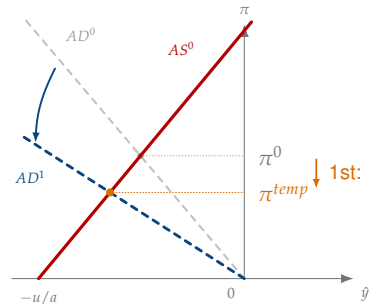
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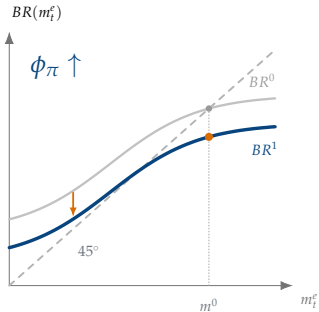
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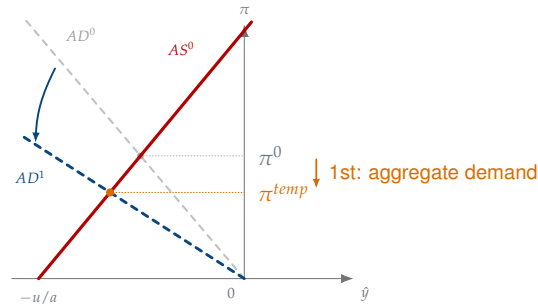
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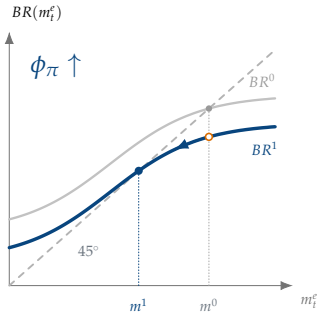
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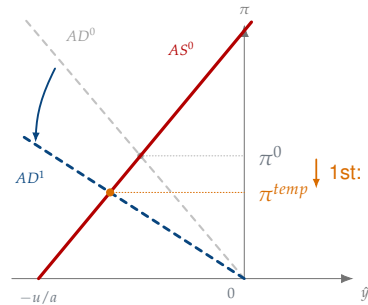
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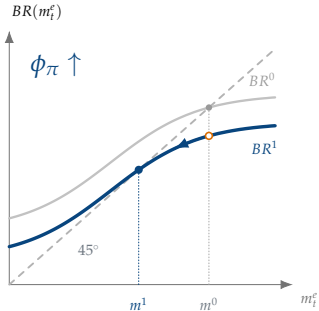
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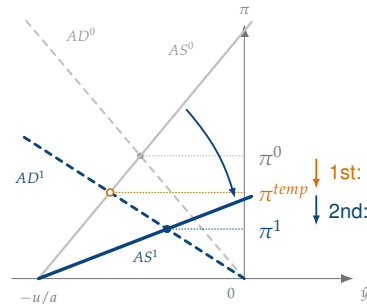
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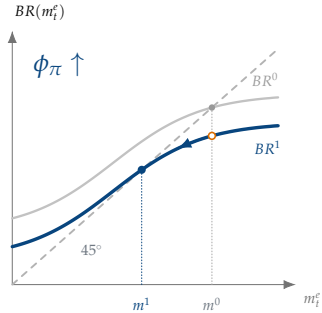
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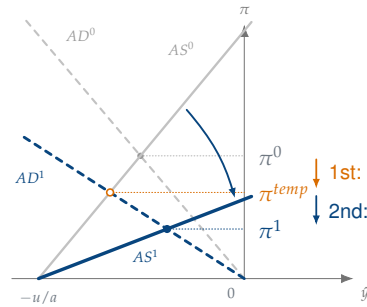
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Can policy ensure uniqueness? (**next**)

▸ Conditions

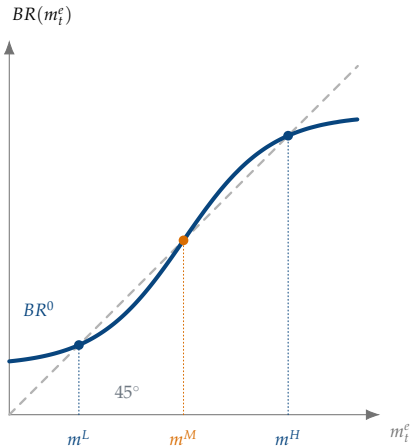
Monetary Policy Can Ensure a Unique Contract Regime

Proposition 5

Under suitable conditions, a stronger policy response ϕ_π *weakens the strategic complementarity:*

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Moreover, there is a unique contracting equilibrium when $\phi_\pi > \phi_\pi^U$ *for some finite* ϕ_π^U.



▸ Conditions

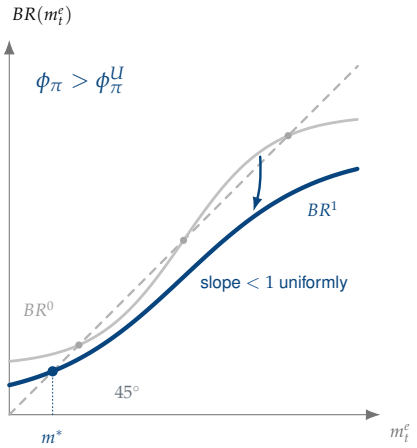
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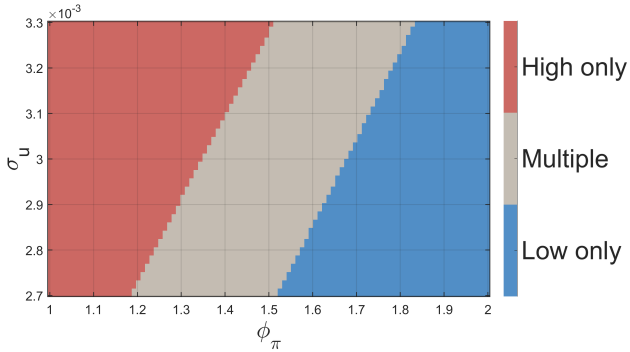
$$\frac{\partial^2 \mathcal{V}}{\partial \phi_\pi \partial m} < 0$$

Moreover, there is a unique contracting equilibrium when $\phi_\pi > \phi_\pi^U$ *for some finite* ϕ_π^U*.*



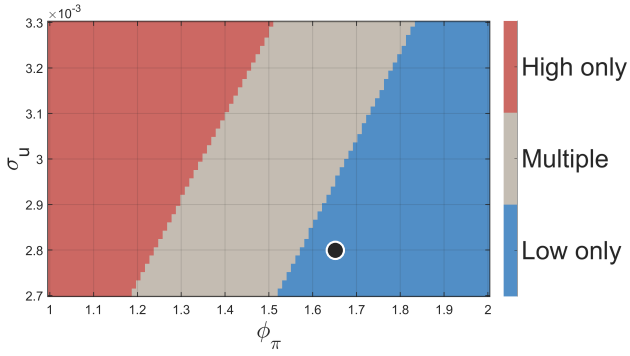
▸ Conditions

Greater Uncertainty Requires More Hawkish Policy



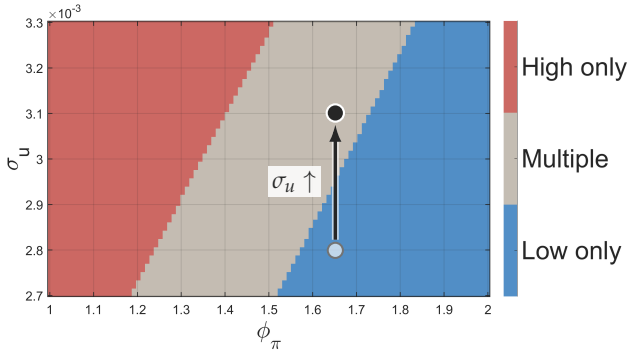
- Higher uncertainty raises the strength of policy response needed for regime anchoring

Greater Uncertainty Requires More Hawkish Policy



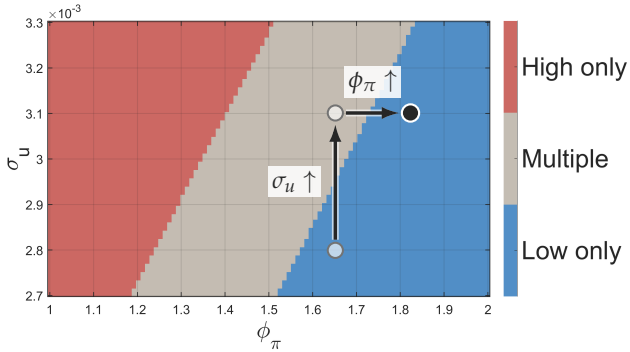
- Higher uncertainty raises the strength of policy response needed for regime anchoring

Greater Uncertainty Requires More Hawkish Policy



- Higher uncertainty raises the strength of policy response needed for regime anchoring

Greater Uncertainty Requires More Hawkish Policy



- Higher uncertainty raises the strength of policy response needed for regime anchoring
- Monetary policy that ensures conditional GE uniqueness may still allow regime multiplicity

Taking Stock of the Theoretical Model

1. Private contract choice: COLA improves private risk-sharing
2. Two nested spirals: COLA amplifies shocks and reinforces adoption
3. Monetary policy: Policy stabilizes inflation by shaping contract choices

Taking Stock of the Theoretical Model

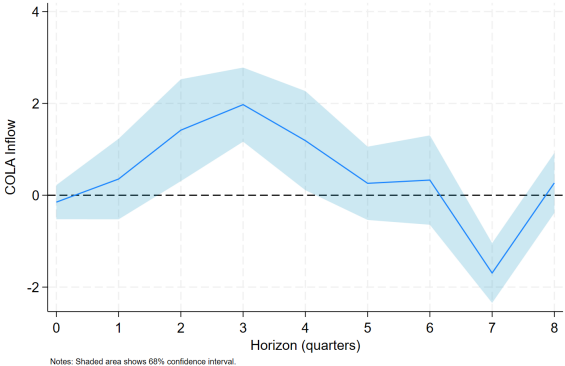
1. Private contract choice: COLA improves private risk-sharing
2. Two nested spirals: COLA amplifies shocks and reinforces adoption
3. Monetary policy: Policy stabilizes inflation by shaping contract choices

- **Micro risk-sharing shapes macro dynamics and policy effectiveness**

Model Validation

Evidence 1: Uncertainty Shocks Raise COLA Adoption

$$COLAIn_{t+h} = \delta_h u_t + \sum_{j=1}^{l} \Gamma_{h,j} Z_{t-j} + \varepsilon_{t+h}$$



Notes: COLA inflow is the fraction of newly renewed contracts that contain COLA provisions. Uncertainty shock u_t is constructed by residualizing the Jurado-Ludvigson-Ng 1-year ahead macroeconomic uncertainty index to its own lags and other macro controls, including inflation, output gap, and FFR.

▸ Subjective inflation uncertainty

Evidence 2: Strategic Complementarity in Contract Choice

Table: Strategic Complementarity in COLA Adoption

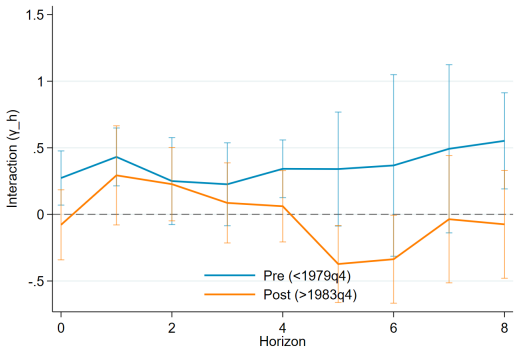
$$COLA_{it} = \beta m_{t-1} + \nu_i + \Gamma' X_{it} + \Lambda' Z_t + \varepsilon_{it}$$

COLA	(1) OLS	(2) JLN IV	(3) Oil IV
COLA share	0.330***	0.458***	0.511***
	(0.056)	(0.141)	(0.141)
Pair FE	✓	✓	✓
Observations	18642	15467	14248

Notes: The dependent variable is an indicator for COLA adoption at renegotiation. COLA share is the one-period lagged COLA share. Column 1 estimates the relationship by OLS. Column 2 instruments lagged COLA share with lags 8--12 of residualized JLN uncertainty shocks, and column 3 instruments lagged COLA share with lags 8--12 of Baumeister-Hamilton oil supply shocks. Macro controls include lags 1--4 of JLN uncertainty, the federal funds rate, and the detrended log real GDP, in addition to log covered workers and contract duration. All specifications include bargaining-pair fixed effects. Standard errors are two-way clustered by bargaining pair and negotiation quarter. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Evidence 3: COLA Amplifies Inflation Response to MP Shocks

$$\pi_{t+h} = \beta_h RR_t + \gamma_h RR_t \times \mathbb{I}_{t-1}^{COLAHigh} + \sum_{j=1}^{l} \Gamma_{h,j} Z_{t-j} + \varepsilon_{t+h}$$



Notes: Error bars are 90% CI. Newey-West HAC lag = h+4.

Notes: This figure shows the coefficient γ_h of the interaction term for different horizons $h = 0, 1, ..., 8$. RR_t refers to Romer-Romer shocks. $\mathbb{I}_{t-1}^{COLAHigh} = 1$ if the one-period lagged COLA share is higher than the median of the subsample in that quarter. Inflation, output gap, and FFR up to 4 lags are controlled.

Calibration and Quantitative Exercise

Quantitative Analysis in the Full Model

- Persistent AR(1) shocks
- 8-quarter staggered contracts

Deterministic paths of shock volatility and contract distribution; realized shocks remain stochastic

Quantitative Questions

Great Inflation (GI, '65-'84)

How much did the contract accelerator amplify inflation?

Great Moderation (GM, '84-'07)

How much did the contract accelerator add to post-Volcker stabilization?

▸ details ▸ stochastic contract distribution

Quantitative Analysis in the Full Model

- Persistent AR(1) shocks
- 8-quarter staggered contracts

Deterministic paths of shock volatility and contract distribution; realized shocks remain stochastic

Two-step Calibration

1 Conditional GE

condition on observed COLA shares

$\bar{m}_{GI} = 0.50, \qquad \bar{m}_{GM} = 0.15$

discipline all non-cost parameters

unique eq'm given contract composition

$\Longrightarrow$

2 Contract Consistency

given non-cost parameters

choose COLA cost structure

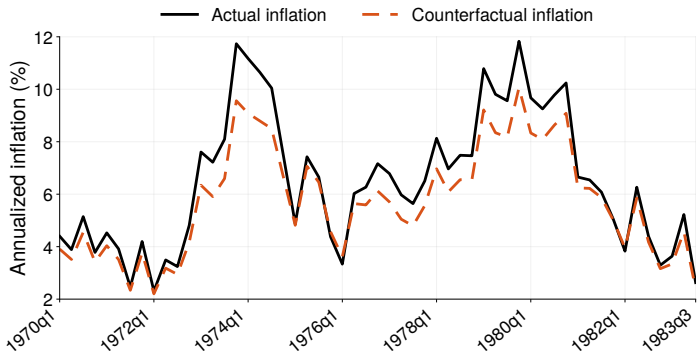
to rationalize $\bar{m}_{GI}, \bar{m}_{GM}$

two shares pin down location and scale

▸ details ▸ stochastic contract distribution ▸ calibrated parameters ▸ targeted moments ▸ IRF

Great Inflation: The Contract Accelerator Amplifies Inflation

Figure: Path of Inflation, actual vs. no COLA response

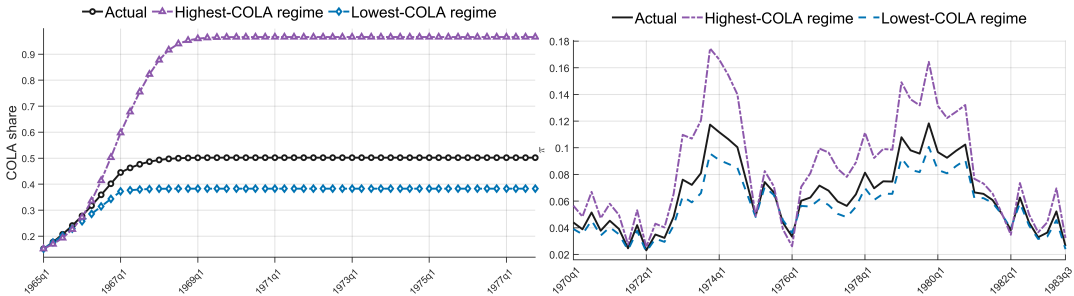


Notes: The Great Inflation is modeled as a permanent unexpected volatility shock to the calibrated level. The black solid line is the actual annualized inflation in the US, and the red dashed line is the counterfactual path of inflation where COLA composition is fixed at the initial level. For the two inflation spikes, endogenous contract amplifies inflation by 14% on average.

▸ different beliefs ▸ robustness

Other Equilibrium Paths during the Great Inflation

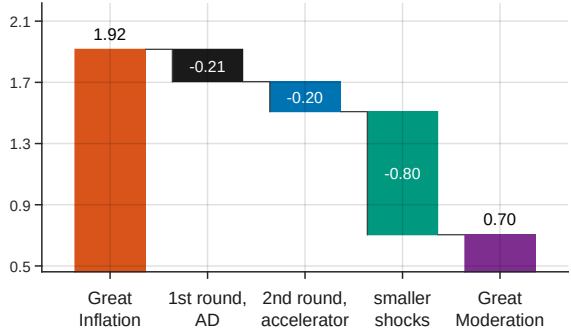
Figure: Paths of the COLA share and inflation in different dynamic equilibria during the Great Inflation



Notes: The left panel plots the equilibrium dynamic path of COLA shares after the GI volatility shock, given the same structural parameters, shocks (both the volatility shock and the inferred markup and demand shocks), and the initial COLA share. They only differ in the initial conjecture about the COLA adoption rate when I solve the model. The black line marked by circles is the actual path, the blue line marked by diamonds is the path when initial conjecture are that no one uses COLA in the future, and the purple line marked by triangles is the path when initial conjecture are that everyone else would always use COLA in the future. The right panel plots the paths of inflation implied by the three scenarios.

Great Moderation: Policy Stabilizes Inflation Twice

Figure: Decomposing the decline in on-impact inflation IRF to a one-std markup shock in each era



Notes: The Great Moderation is modeled as a more hawkish rule and a decline in shock volatilities. This figure decomposes the decline in on-impact inflation response to the respective one-std markup shocks from GI to GM. Numbers are annualized percentage points. "AD" means the effect of monetary policy through the aggregate demand channel, "accelerator" is the effect of monetary policy through the contract choice channel where COLA is endogenously chosen, but primitive-shock volatility remains at the GI level, and "smaller shocks" means the effect from the exogenous decline in shock volatility.

▸ details ▸ Demand shocks ▸ Decomposing volatilities ▸ Welfare implications

Other Exercises, Robustness Checks, and Extensions

- Robustness for different extrapolations from union to non-union detail
- Decomposing the decline in unconditional std of inflation detail
- Welfare tension: private insurance vs. aggregate externality in inflation risk detail
- Optimal COLA-augmented Taylor rule detail
- Endogenous stochastic inflation volatility detail

Conclusions

- **Theory:** wage-setting arrangements are endogenous
  - risk sharing makes wage rigidity state-dependent
  - gives rise to the contract accelerator
- **Monetary policy:** anchors the indexation regime
  - lowers COLA adoption and weakens propagation
  - must account for uncertainty and beliefs
- **Quantification:** contract choice matters in practice
  - GI: endogenous adoption amplifies inflation by 14%
  - GM: contract choice doubles policy's contribution

Micro risk-sharing shapes macro dynamics and policy effectiveness

COLA in Non-profit Firms

(a) COLAs will be allocated to both full-time and part-time employees, ensuring equitable treatment. Seasonal employees, however, will instead receive minimum wage increases starting in January each year.

The adjustment will be based on the Consumer Price Index (CPI) for All Urban Consumers within specific size classifications (B/C). The CPI will be assessed from April of the preceding year to March of the current year. Should significant economic changes occur in May ahead of the budget approval, a recalculation of the COLA may take place. In the absence of such changes, the initial COLA rate calculated in April will be proposed in the final budget.

(b) All eligible employees, whether full-time or part-time, will receive the same dollar amount as their COLA.

Figure: COLA in Washington Association of Conservation Districts, 2025 Employment Handbook

back

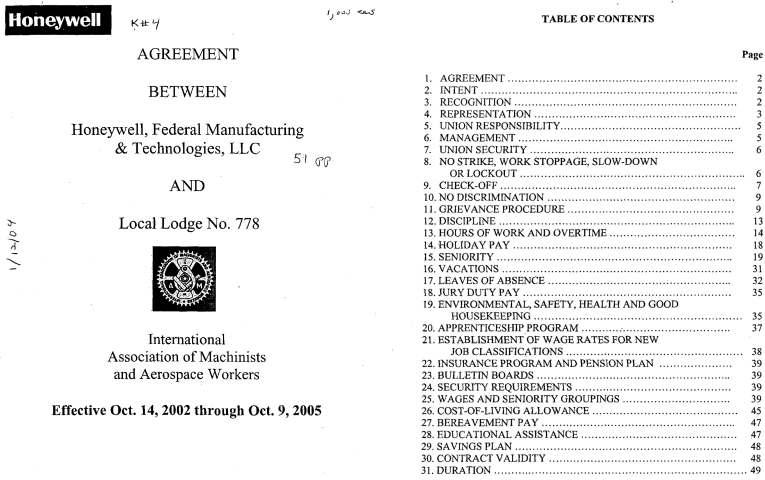
COLA IN CEO AGREEMENTS

During the Senior Status Period, the Executive shall be entitled to an annual salary (the “Senior Status Salary”), payable in accordance with the regular payroll practices of the Company, of ... $400,000 plus the COLA Adjustment ... which reflects any increase in the cost of living on the immediately preceding June 30th over the cost of living on June 30, 2000, using as a basis for such increase the Consumer Price Index for all Urban Consumers (CPI-U) for New York, Northern New Jersey-Long Island, as published by the U.S. Department of Labor (the “Index”) .

Employment Agreement between Bed Bath & Beyond Inc. and Warren Eisenberg (the “Executive”).

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Example Collective Bargaining Contract Files



Honeywell

K# 4

1,000 emp

AGREEMENT

BETWEEN

Honeywell, Federal Manufacturing
& Technologies, LLC

51 pp

AND

Local Lodge No. 778

1/1/10 4

International
Association of Machinists
and Aerospace Workers

Effective Oct. 14, 2002 through Oct. 9, 2005

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Summary Statistics

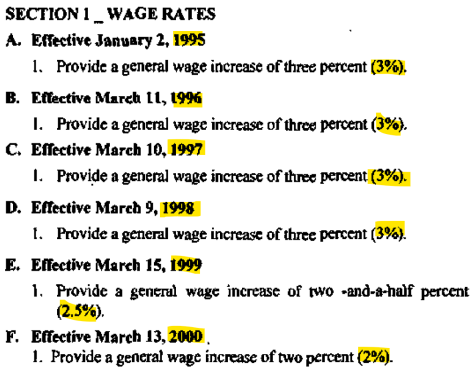
Table: Summary Statistics

	N	Mean	SD	P25	P50	P75
Panel A: Full sample						
Contract duration (months)	21,543	28.8	11.0	24	36	36
Union size	21,543	4,190.6	14,109.0	1,240	1,850	3,500
Scheduled wage change (=1)	21,543	0.93	0.26	1	1	1
COLA (=1)	21,543	0.22	0.42	0	0	0
Panel B: Conditional on COLA						
COLA duration (months)	4,676	20.2	11.3	12	21	30
Union size	4,798	6,937.3	27,229.2	1,300	2,100	4,400
COLA pass-through	3,984	0.89	0.42	0.72	0.86	1

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Actual Rules in Wage Contracts

Figure: Examples of scheduled increases



SECTION 1 _ WAGE RATES

A. Effective January 2, 1995

1. Provide a general wage increase of three percent (3%).

B. Effective March 11, 1996

1. Provide a general wage increase of three percent (3%).

C. Effective March 10, 1997

1. Provide a general wage increase of three percent (3%).

D. Effective March 9, 1998

1. Provide a general wage increase of three percent (3%).

E. Effective March 15, 1999

1. Provide a general wage increase of two -and-a-half percent (2.5%).

F. Effective March 13, 2000

1. Provide a general wage increase of two percent (2%).

back

Actual Rules in Wage Contracts

Figure: Examples of COLA (right)

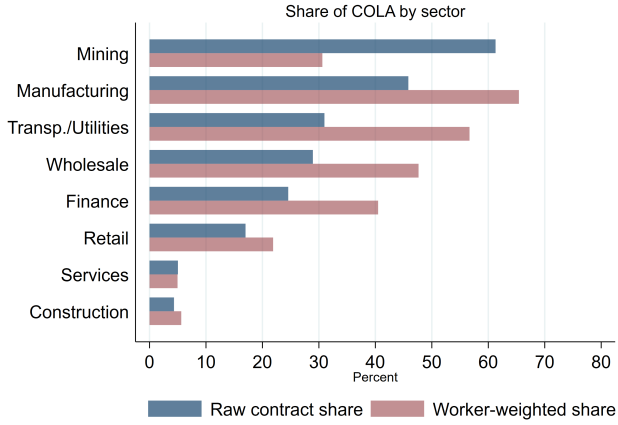
In determining the three-month average of the Index for a specified period, the computed average shall be rounded to the nearest one-tenth (0.1) Index point. The amount of the Cost-of-Living Allowance shall be adjusted on the basis of one cent ($.01) per hour for each four-tenths (0.4) point change in the three-month average of the Indexes. The amount of Cost-of-Living Allowance shall be in accordance with the following table, subject to the restrictions of paragraph C.3.:

Three-Month Average BLS Consumer Index (1967=100)	Cost-of-Living Allowance
524.8-525.1	0.00
525.2-525.5	0.01
525.6-525.9	0.02
526.0-526.3	0.03
526.4-526.7	0.04
526.8-527.1	0.05
527.2-527.5	0.06

and so forth with one-cent ($.01) adjustments for each four-tenths (0.4) point change in the average Index for the appropriate three months as indicated in paragraph C.1.

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COLA Prevalence Within Different Sectors

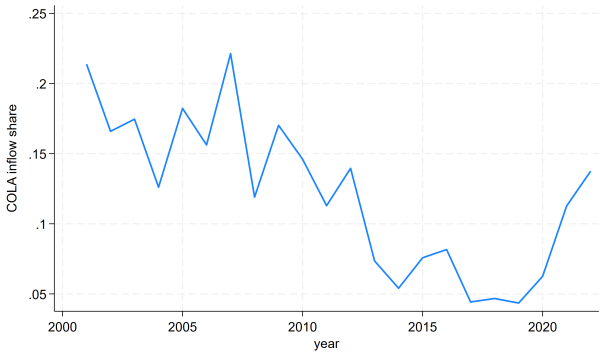


Notes: This figure plots the share of COLA contracts by sector from the main CBA dataset. "Worker-weighted share" means that the share is weighted by the number of workers covered by each contract.

back

Surge of Wage Indexation Post-Pandemic

Figure: Share of new contracts with COLA provision

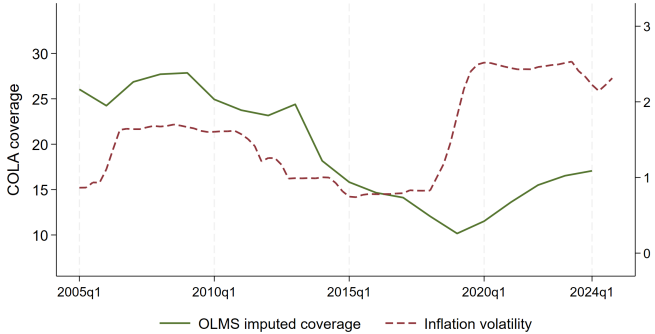


Notes: This figure plots the share of new contracts with COLA provision each year from 2001-2022. The data is digitized from the collective bargaining agreements collected by the Office of Labor Management Standards.

back

Surge of Wage Indexation Post-Pandemic

Figure: Extended COLA coverage against inflation volatility

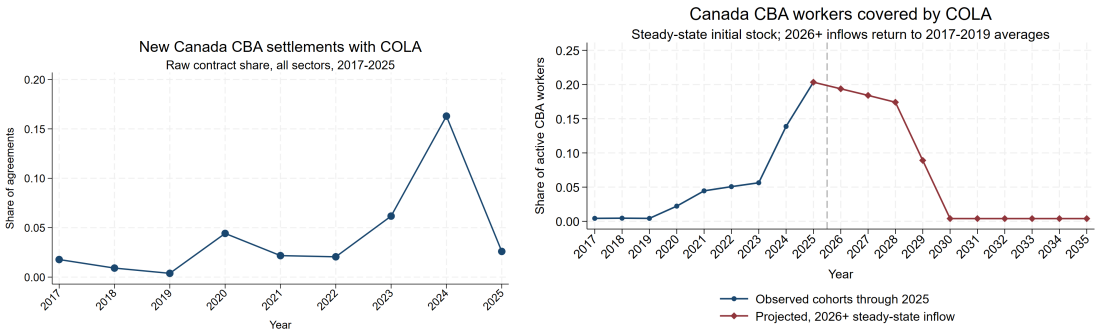


Notes: The figure plots COLA coverage in the US against inflation volatility. BLS coverage means the series is from the main 1968-1995 dataset, and the OLMS imputed coverage is from the OLMS collected collective bargaining agreements from 2001 to 2022.

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Surge of Wage Indexation In Canada

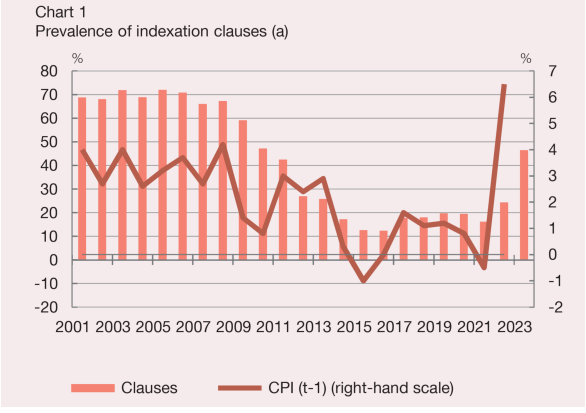
Figure: Share of new contracts with COLA provision and the imputed coverage



Notes: The left panel plots the time path of the share of new collective bargaining contracts with COLA clauses in Canada from 2017 to 2025, while the right panel plots the imputed worker coverage from the inflow sequence. The red part in the right panel is the projection of COLA coverage from 2026 to 2035, assuming that COLA inflow goes back to the 2017-2019 average from 2026 on.

back

Surge of Wage Indexation In Spain



Notes: The figure plots the fraction of new collective bargaining contracts with COLA clauses each year (the pink bar) against aggregate inflation (the red curve) in Spain from 2001 to 2023. This figure comes directly from Spain's central bank report (Economic Bulletin, 3/2022, Banco De Espana)

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Aggregate State-Dependent Nominal Pass-Through

Table: Job-stayer wage pass-through to inflation

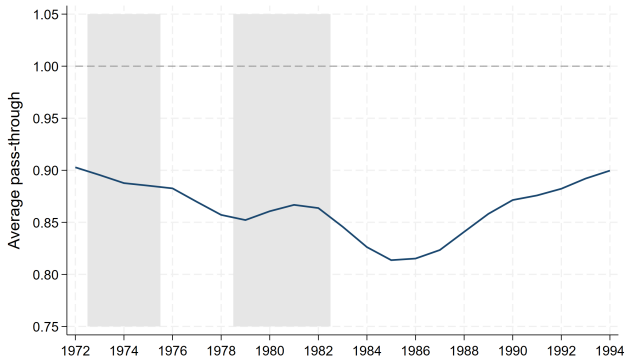
$$g_t^{W,stayer} = \beta \pi_t^{12m} + \gamma \left(\pi_t^{12m} \times \text{Spike}_t \right) + \delta_{y(t)} + \varepsilon_t$$

Stayer wage growth	(1) OLS	(2) Positive WTI IV
12-month CPI inflation	-0.007	-0.000
	(0.038)	(0.080)
Inflation × Spike	0.176***	0.218**
	(0.061)	(0.094)
Observations	310	300
Year FE	Yes	Yes

Notes: The dependent variable is the Atlanta Fed Wage Growth Tracker measure of year-over-year median nominal wage growth for job stayers. The inflation variable is same-window twelve-month CPI inflation. The Spike indicator equals one during 2021-2022, and the reported interaction coefficient measures the increase in inflation pass-through relative to the pre-2021 period. Year fixed effects are partialled out in all specifications. Standard errors are HAC/Newey-West with a Bartlett kernel and twelve monthly lags. In column (2), twelve-month CPI inflation and its interaction with the spike indicator are instrumented with the cumulative positive WTI price changes over months $t-1$ through $t-12$. back

Average COLA Pass-Through of All Active Contracts

Figure: Average pass-through of *active* COLA contracts each year

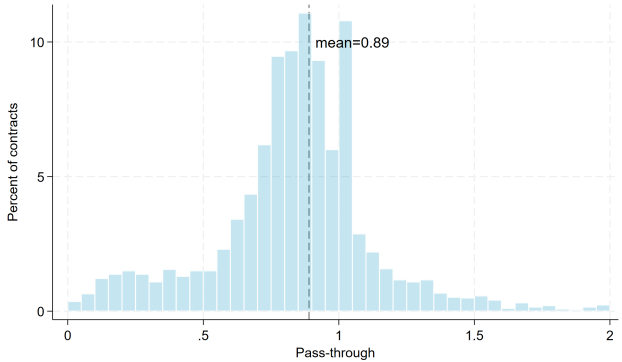


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Cross-Sectional Distribution of COLA Pass-Through

- Price to nominal wage pass-through

$$\Pi_{it} = \frac{d \log W_{it}}{d \log P_t}$$



back

Substitution between the Two Rules

- Scheduled increases are universal but generally lower when COLA is provided

$$SchedIncrease_{it} = \beta COLA_{it} + \Gamma Z_{it} + \delta_{y(t)} + \nu_i + \varepsilon_{it}$$

	OLS (1)	OLS (2)	2SLS (3)
COLA	-0.801	-1.121	-6.734
	(0.085)	(0.215)	(3.279)
Pair FE	✓	✓	✓
Financial controls		✓	✓
N	20,790	4,537	3,394

Notes: Standard errors clustered at year and firm-union pair level. The first regression is on the whole sample, the last two are on the Compustat-linked sample with financial controls. For the 2SLS one, $COLA_{it}$ is instrumented by Baumeister-Hamilton oil supply shocks 1 and 2 quarters ago interacted with firm fixed assets, and policy rate (FFR) is also controlled.

back

Formal Equilibrium Definition

- A *rational expectations contracting general equilibrium* is a collection of stochastic processes

$$\left\{\left\{L_{it}, P_{it}^{X}, \mathbb{I}_{it}^{COLA}\right\}_{i\in[0,1]}, \left\{P_{ft}, P_{ft}^{*}, Y_{ft}\right\}_{f\in[0,1]}, \left\{W_{t,k}^{COLA} \in \mathcal{C}_{t-k}^{c}, W_{t,k}^{sched} \in \mathcal{C}_{t-k}^{s}\right\}_{k=1}^{N}, \mathcal{M}_t, X_t, Y_t, C_t^e, P_t, \pi_t, B_{t+1}, \mathcal{R}_t, i_t, \rho_t, \varepsilon_{yt}\right\}_{t\in\mathbb{Z}}$$

  such that,

  1. Each service firm-union pair chooses contract type, wage rule, and labor input optimally
  2. Goods firms choose price P_{ft}^{*} if they can reset; entrepreneurs choose C_t^e optimally.
  3. All agents have expectations that are consistent with the equilibrium laws of motion conditional on their respective information sets.
  4. The demand shock ρ_t and the markup shock u_t evolve exogenously; The contract distribution $\mathcal{M}_t$ evolves endogenously through firms' contract choice.
  5. All markets clear.
- A *conditional equilibrium* is defined similarly but with $\{\mathcal{M}_t\}_{t\in\mathbb{Z}}$ exogenously given back

Details of the Price to Real Wage Pass-through $K(m_t)$

- From Lemma 1 and the guessed process of π_t and $\hat{\zeta}_t$

$$\hat{w}_t^{COLA}(\pi_t) = \mu \frac{\zeta_\rho(m_t)\pi_\rho(m_t)\sigma_\rho^2 + \zeta_u(m_t)\pi_u(m_t)\sigma_u^2}{\pi_\rho(m_t)^2\sigma_\rho^2 + \pi_u(m_t)^2\sigma_u^2}\pi_t$$

- Therefore

$$\hat{\omega}_t := m_t\hat{w}_t^{COLA} + (1-m_t)\hat{w}_t^{sched} = \underbrace{\left[\mu m_t \left(\frac{\zeta_\rho(m_t)\pi_\rho(m_t)\sigma_\rho^2 + \zeta_u(m_t)\pi_u(m_t)\sigma_u^2}{\pi_\rho(m_t)^2\sigma_\rho^2 + \pi_u(m_t)^2\sigma_u^2}\right) - (1-m_t)\right]}_{:=\mathcal{K}(m_t)}\pi_t$$

back

Lemma 3

Given Assumption 1, the conditional equilibrium for any given COLA share m_t is characterized by

$$\pi_t = \kappa\left(\frac{\varepsilon_x - \iota_1}{\iota_1 \varepsilon_x}\hat{y}_t + \frac{\iota_2}{\iota_1}\hat{\omega}_t\right) + \kappa u_t$$

$$\hat{\omega}_t = \mu m_t \left(\frac{\zeta_\rho(m_t)\pi_\rho(m_t)\sigma_\rho^2 + \zeta_u(m_t)\pi_u(m_t)\sigma_u^2}{\pi_\rho(m_t)^2\sigma_\rho^2 + \pi_u(m_t)^2\sigma_u^2}\right)\pi_t - (1-m_t)\pi_t$$

$$\hat{y}_t = (\iota_1\hat{\zeta}_t - \iota_2\hat{\omega}_t)$$

$$\hat{y}_t = \frac{\bar{C}^e}{\bar{Y}}\hat{c}_t^e + \varphi_\omega\hat{\omega}_t + \varphi_\zeta\hat{\zeta}_t$$

$$\hat{c}_t^e = -\gamma(\phi_\pi\pi_t + \phi_y\hat{y}_t - \hat{\rho}_t)$$

back

Regularity Conditions

Define

$$\Omega := \left(1 + \frac{\bar{C}}{\bar{Y}}\sigma\phi_y\right)^{-1},\ \Lambda_\sigma := \frac{\bar{C}}{\bar{Y}}\sigma,\ \Lambda_\pi := \Lambda_\sigma\phi_\pi$$

$$C_\zeta^S := \kappa\left(1 - \frac{\iota_1}{\varepsilon_x}\right),\ C_\zeta^D := \iota_1 - \Omega\varphi_z,\ A := \iota_2 + \Omega\varphi_\omega,\ B := \kappa\frac{\iota_2}{\varepsilon_x},$$

$$D_0 := (\Omega\Lambda_\sigma\sigma_\rho)^2(C_\zeta^S)^2 + \sigma_u^2(C_\zeta^D)^2,\ N_0 := (\Omega\Lambda_\sigma\sigma_\rho)^2C_\zeta^S - \sigma_u^2C_\zeta^D\Omega\Lambda_\pi,\ N_1 := -(\Omega\Lambda_\sigma\sigma_\rho)^2C_\zeta^SB + \sigma_u^2C_\zeta^DA.$$

For a given $m_t \in [0, 1]$, write

$$\begin{aligned}
Q(m_t) &:= D_0 - \mu m_t N_1,\\
R(m_t) &:= \mu m_t N_0 - D_0(1 - m_t),\\
\tilde{\Delta}(m_t) &:= C_\zeta^D\left[Q(m_t) - BR(m_t)\right] - C_\zeta^S\left[AR(m_t) - \Omega\Lambda_\pi Q(m_t)\right],\\
\Pi(m_t) &:= \frac{D_0 + \mu(N_0 - N_1)}{Q(m_t)},\\
G &:= BC_\zeta^D + C_\zeta^SA.
\end{aligned}$$

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Regularity Conditions

The regularity conditions include

$$C_\zeta^D > 0, \qquad C_\zeta^S > 0, \qquad A > 0, \qquad B > 0$$

In addition,

$$N_1 > 0, \qquad D_0 - \mu N_1 > 0, \qquad \widetilde{\Delta}(0) > 0, \qquad \widetilde{\Delta}(1) > 0$$

and

$$0 < D_0 + \mu(N_0 - N_1) < D_0 - \mu N_1$$

In general, these assumptions are not restrictive. Loosely speaking, one can think of them as the minimal set of assumptions to guarantee that the price-to-nominal-wage pass-through is between zero and one under COLA contracts

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Proposition 6

Using the notations defined above, the conditional GE is unique, and

$$\pi_u(m_t) = \frac{C_\zeta^D Q(m_t)}{\widetilde{\Delta}(m_t)}, \qquad \zeta_u(m_t) = \frac{AR(m_t) - \Omega\Lambda_\pi Q(m_t)}{\widetilde{\Delta}(m_t)},$$

$$\pi_\rho(m_t) = \frac{\Omega\Lambda_\sigma C_\zeta^S Q(m_t)}{\widetilde{\Delta}(m_t)}, \qquad \zeta_\rho(m_t) = \frac{\Omega\Lambda_\sigma \left[Q(m_t) - BR(m_t)\right]}{\widetilde{\Delta}(m_t)}.$$

Moreover,

$$\pi_u(m_t) > 0, \qquad \pi_\rho(m_t) > 0, \qquad \zeta_\rho(m_t) > 0, \qquad \zeta_u(m_t) < 0,$$

and the nominal COLA pass-through is partial:

$$0 < \Pi(m_t) < 1.$$

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THE FULL MODEL

- The full model focuses on deterministic time path of volatilities of primitive shocks:

$$\left\{\Sigma_t : \Sigma_t = diag(\sigma_{ut}^2, \sigma_{\rho t}^2)\right\}_{t\in\mathbb{Z}} \text{ is deterministic}$$

- With AR(1) shocks and N-period contracts, π_t and $\hat{\zeta}_t$ follow an MA$(N+1)$ representation:

$$\pi_t = \sum_{s=0}^{N} q_{t,s}^{\pi} v_{t-s}, \qquad \hat{\zeta}_t = \sum_{s=0}^{N} q_{t,s}^{\zeta} v_{t-s}$$

  where $v_t = (u_t, \rho_t)'$

- Given an initial deterministic guess of the contract distribution, the implied gain from COLA, and therefore, the dynamics of the contract distribution, are deterministic

$$\mathcal{M}_{t,1} = \frac{1}{N} F_{\chi}\left(\mathcal{V}_{t-1}\left(\{\widetilde{\mathcal{M}}_{t+\tau}\}_{\tau\geq 0}; \{\Sigma_{t+\tau}\}_{\tau\geq 0}\right)\right), \qquad \mathcal{M}_{t,k} = \mathcal{M}_{t-1,k-1}, \quad k = 2, \ldots, N$$

- A separation of first-order shocks from the contract distribution

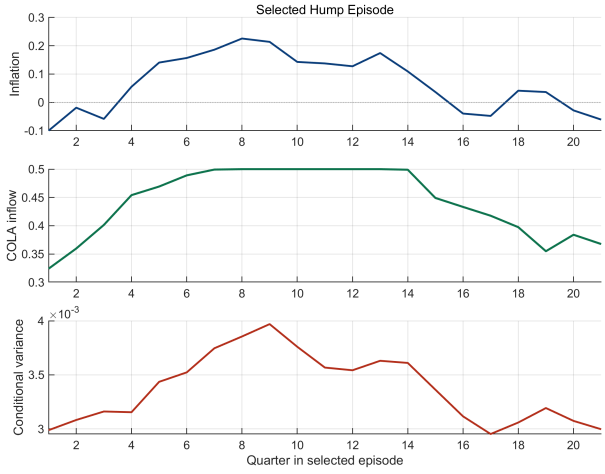
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Stochastic Contract Distribution and Inflation Volatilities

- A separation of first-order shocks from the contract distribution in the baseline, because
  - the gain from COLA depends only on conditional second-order moments
  - the assumption of deterministic time path of primitive volatility
- One departure: level of inflation enters the gain from COLA
  - e.g., higher inflation forecast error raises fear for the future and therefore the value of COLA
  - implications: endogenous stochastic inflation volatility through the contract choice channel even if primitive volatility is constant
- Computational difficulty: the $MA(N+1)$ form fails; nonlinear forecast problem
  - parametrized expectations in the state space

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Illustration Using a Two-Period Contract Case



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Higher Subjective Inflation Uncertainty Raises COLA Adoption

Table: COLA adoption and subjective inflation uncertainty

$$COLA_{it} = \beta U_t + \theta \pi_t + \nu_i + \Gamma' X_{it} + \Lambda' Z_t + \varepsilon_{it}$$

COLA adoption	(1) IQR, 2SLS	(2) Var, 2SLS
Subjective inflation uncertainty, IQR	0.084**	
	(0.036)	
Subjective inflation uncertainty, variance		0.009**
		(0.004)
Pair FE	✓	✓
Macro controls	✓	✓
Observations	14248	14248

Notes: The dependent variable is an indicator for COLA adoption at renegotiation. Subjective inflation uncertainty is constructed from Michigan Survey one-year-ahead inflation forecasts. The IQR specification uses the weighted monthly interquartile range of individual forecasts after normalizing each forecast by the monthly mean forecast; the variance specification uses the corresponding weighted cross-sectional variance. Inflation and subjective uncertainty are instrumented with lags 1--4 of Romer-Romer monetary policy shocks and lags 1--4 of residualized JLN uncertainty shocks. Macro controls include lags 1--4 of the federal funds rate, lagged detrended log employment, and lags of detrended log real GDP, in addition to log covered workers and contract duration.

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Externally Calibrated Parameters

Table: Externally calibrated parameters

Parameter	Value	Meaning
Panel A: Externally calibrated parameters		
β	0.99	Discount factor (quarterly)
γ	3	Worker risk aversion
ε	5	Elasticity of substitution
θ	0.75	Probability of not resetting prices
N	8	Contract length
σ_ξ	0.25	Std. dev. of firm idiosyncratic shock
$\phi_{\pi,GI}$	1.1	MP response to inflation, GI
$\phi_{\pi,GM}$	1.7	MP response to inflation, GM
ϕ_y	0.4	MP response to output gap

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Internally Calibrated Parameters

Table: Internally calibrated parameters

Parameter	Value	Meaning
Panel B: Internally calibrated parameters		
ϕ_0	0.3027	Slope of the external financing cost
ν	0.0093	Slope of the labor adjustment cost
b	0.5087	Worker home production
ϱ_u	0.8487	AR(1) persistence of markup shocks
ϱ_ρ	0.8358	AR(1) persistence of demand shocks
$\sigma_{u,GI}$	0.0013	Std. of innovation to markup shocks, GI
$\sigma_{\rho,GI}$	0.005	Std. of innovation to demand shocks, GI
$\sigma_{u,GM}$	0.0007	Std. of innovation to markup shocks, GM
$\sigma_{\rho,GM}$	0.0019	Std. of innovation to demand shocks, GM
χ_0	0.0054	COLA cost distribution location parameter
χ_1	0.0028	COLA cost distribution scale parameter

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Targeted Moments

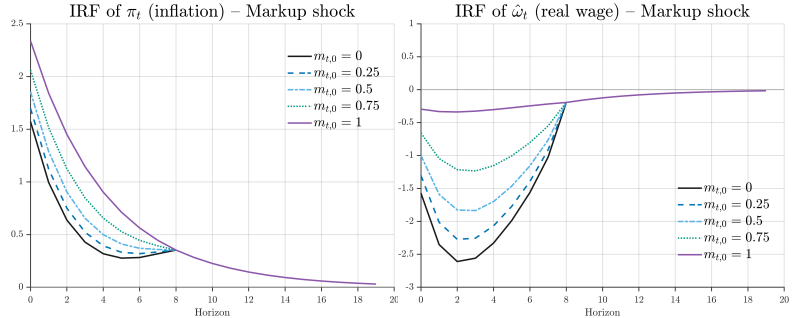
Table: Targeted moments

Targeted moments	Data values	Model values
Value of cash	1.13	1.15
Labor share	0.60	0.66
COLA pass-through	0.87	0.88
ACF(1) of real GDP	0.85	0.84
ACF(1) of inflation	0.85	0.88
Std. dev. of real GDP during GI	0.020	0.020
Std. dev. of inflation during GI	0.008	0.009
Std. dev. of real GDP during GM	0.009	0.010
Std. dev. of inflation during GM	0.003	0.004
Average COLA stock during GI	0.50	0.51
Average COLA stock during GM	0.15	0.15

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Inflation IRF to Markup Shocks under Different COLA shares

Figure: IRFs of inflation and real wage after a markup shock

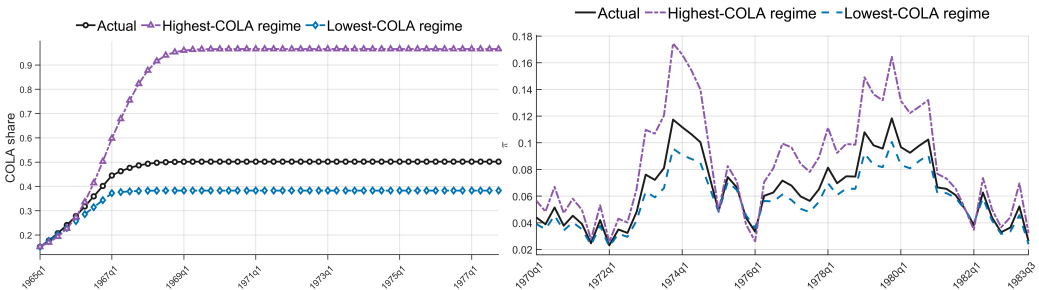


*Notes:*The left (right) panel plots the impulse response functions (IRFs) of inflation (aggregate real wage) with respect to a unit markup shock in the stationary conditional general equilibrium with different COLA shares.

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What if COLA Adoption Had Been Different?

Figure: COLA and inflation paths after the GI shock in different dynamic equilibria

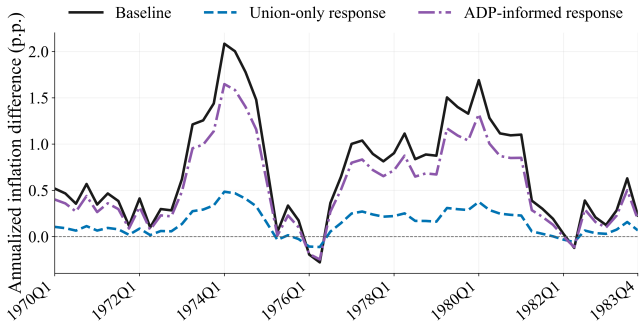


Notes: The left panel plots the equilibrium dynamic path of COLA shares after the GI volatility shock from different initial conjecture about the COLA adoption rate. The black line marked by circles is the actual path, the blue line marked by diamonds is the path when initial conjecture are that no one uses COLA in the future, and the purple line marked by triangles is the path when initial conjecture are that everyone else would always use COLA in the future. The right panel plots the paths of inflation implied by the three scenarios.

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Different Extrapolations from Union to the Broader Economy

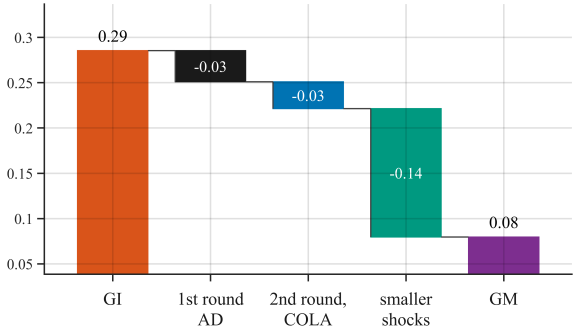
Figure: Amplification during the Great Inflation under different extrapolation assumptions



Notes: Baseline extrapolation assumes the same COLA coverage among union and non-union workers. Union-only response assumes no non-union indexation. ADP-informed response uses the recent inflation surge: U.S. COLA coverage rose by 7 pp, while the share of workers receiving off-cycle raises rose by 10 pp (Hurst et al., 2026). The GI's COLA increase was five times larger (35 pp), so I scale the off-cycle increase by five. I then halve it, assuming two wage adjustments per year versus typically four under quarterly COLA. This gives a 25 pp increase in effective non-union indexation, from 17% to 42%. back

Great Moderation: Policy Stabilizes Inflation Twice

Figure: Decomposing the decline in on-impact inflation IRF to respective one-std demand shock



Notes: The Great Moderation is modeled as a more hawkish rule and a decline in shock volatilities. This figure decomposes the decline in on-impact inflation response to the respective one-std demand shocks from GI to GM. "AD" means the effect of monetary policy through the aggregate demand channel, "COLA" is the effect of monetary policy through the contract choice channel where COLA is endogenously chosen, but primitive-shock volatility remains at the GI level, and "smaller shocks" means the effect from the exogenous decline in shock volatility.

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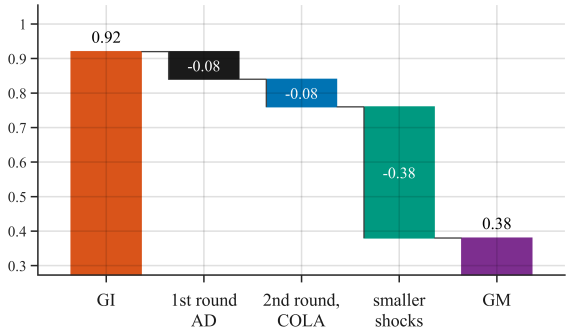
Great Moderation: Decomposing Policy vs. Luck

- GM: modeled as 1) a more hawkish policy rule + 2) a decline in primitive shock volatilities
- Decomposing the total effect as the following steps:
  1. only policy rule changes, fixing COLA share and shock vol. at GI level
  2. +COLA response endogenously to the policy change
  3. +smaller shock volatilities

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Great Moderation: Decomposing the Decline in Inflation Volatilities

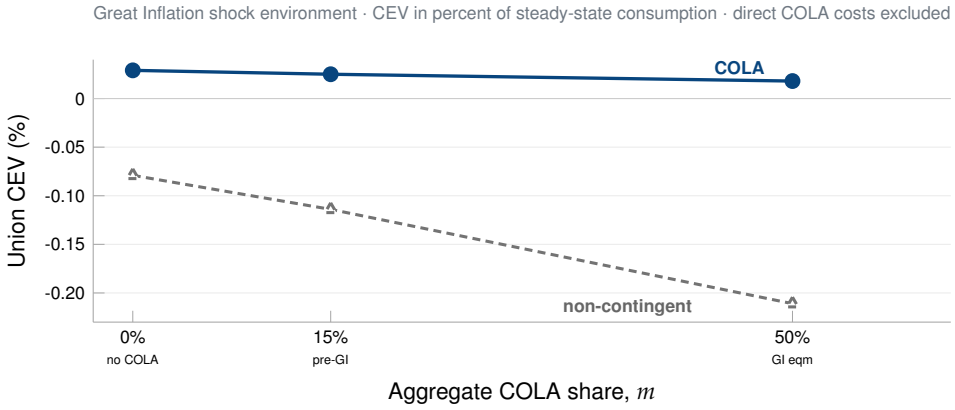
Figure: Decomposing the decline in inflation volatility



Notes: The Great Moderation is modeled as a more hawkish rule and a decline in shock volatilities. This figure decomposes the decline in unconditional standard deviation of inflation from GI to GM. "AD" means the effect of monetary policy through the aggregate demand channel, "COLA" is the effect of monetary policy through the contract choice channel where COLA is endogenously chosen, but primitive-shock volatility remains at the GI level, and "smaller shocks" means the effect from the exogenous decline in shock volatility.

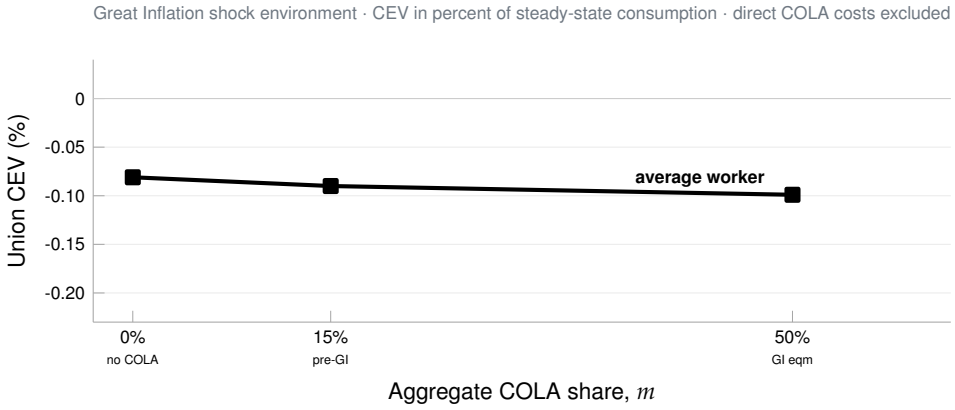
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COLA Insures Adopters—but Lowers Aggregate Welfare



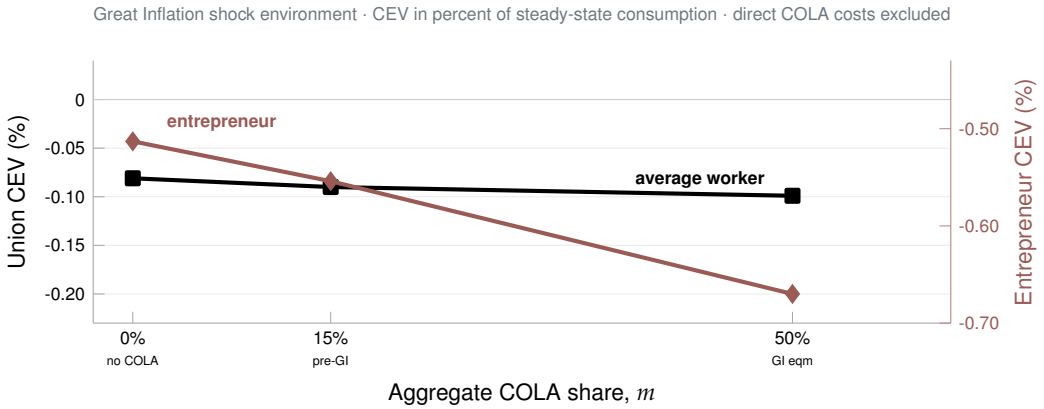
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COLA Insures Adopters—but Lowers Aggregate Welfare



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COLA Insures Adopters—but Lowers Aggregate Welfare



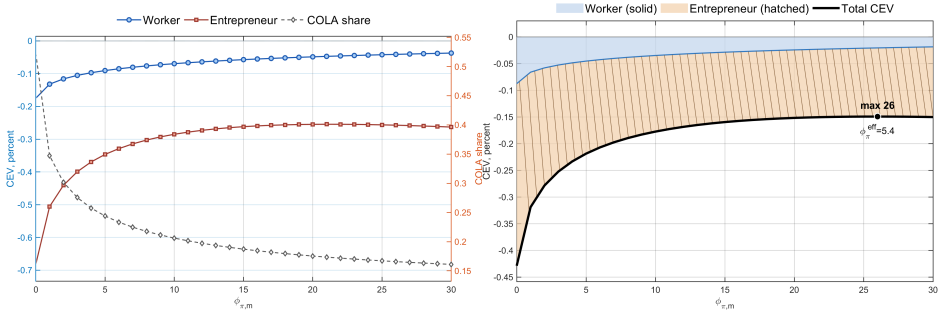
Private insurance creates a general-equilibrium welfare externality

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Optimal COLA-Augmented Taylor Rule

$$i_t = (\phi_{\pi 0} + \phi_{\pi m} m_t)\, \pi_t + \phi_y \hat{y}_t$$

Figure: Welfare under the COLA-augmented Taylor rule



Notes: The figure reports consumption-equivalent welfare for workers, net of COLA adoption costs, and for the entrepreneur under alternative values of$\phi_{\pi m}$.

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